

-- Analyze total passenger traffic per route and over time.

-- top 5 routes which are the busiest routes having huge passenger traffic

/*Miami, FL New York, NY

New York, NY Miami, FL

Orlando, FL New York, NY

New York, NY Orlando, FL

New York, NY Chicago, IL*/

-- Query used

/*select ORIGIN_CITY_NAME, DEST_CITY_NAME, sum(PASSENGERS) as total_passengers

from airport_data

group by 1,2

order by total_passengers desc

limit 5*/

-- Least busiest routes

/*select ORIGIN_CITY_NAME, DEST_CITY_NAME, sum(PASSENGERS) as total_passengers

from airport_data

where ORIGIN_CITY_NAME <> DEST_CITY_NAME

group by 1,2

having sum(PASSENGERS) > 0

order by total_passengers

limit 5 */

--. Determine average passengers per flight for various routes and airports.

-- For routes

/*select ORIGIN_CITY_NAME, DEST_CITY_NAME, avg(PASSENGERS) as avg_passengers

from airport_data

```

group by 1,2
order by avg_passengers desc*/

-- for airports
/*
with outgoing_passengers as
(select ORIGIN_AIRPORT_ID, avg(PASSENGERS) as avg_passengers
from airport_data
group by 1
order by avg_passengers desc),

incoming_passengers as
(select DEST_AIRPORT_ID, avg(PASSENGERS) as avg_passengers
from airport_data
group by 1
order by avg_passengers desc),

all_airports as
(select distinct ORIGIN_AIRPORT_ID as airport_id from airport_data
union
select distinct DEST_AIRPORT_ID as airport_id from airport_data
)

select aa.airport_id, op.avg_passengers + ip.avg_passengers as total_passenger_travelling
from all_airports as aa
left join outgoing_passengers as op on aa.airport_id = op.ORIGIN_AIRPORT_ID
left join incoming_passengers as ip on aa.airport_id = ip.DEST_AIRPORT_ID
where op.avg_passengers + ip.avg_passengers is not null
order by total_passenger_travelling desc*/

-- Assess flight frequency and identify high-traffic corridors.

```

```
/*select ORIGIN_CITY_NAME, DEST_CITY_NAME, count(AIRLINE_ID) as total_flights
from airport_data
group by 1,2
order by total_flights desc
limit 10*/
```

-- Evaluate available seat capacity to understand seat utilization for each airline

```
/*with seating_capacity as
(select AIRLINE_ID, max(PASSENGERS) as seat_capacity
from airport_data
group by 1
having max(PASSENGERS) > 0),
```

seat_utilization as

```
(select ad.AIRLINE_ID, ad.PASSENGERS*100.00/sc.seat_capacity as seat_utilization
from airport_data ad join seating_capacity sc on ad.AIRLINE_ID = sc.AIRLINE_ID
order by seat_utilization desc)
```

```
select AIRLINE_ID, round(avg(seat_utilization),2) as avg_seat_utilization
from seat_utilization
group by 1
order by avg_seat_utilization desc*/
```

/* Identify popular destination airports based on inbound passenger counts.

```
SELECT
DEST_AIRPORT_ID,
DEST_CITY_NAME,
SUM(PASSENGERS) AS inbound_passengers
FROM airport_traffic
```

GROUP BY DEST_AIRPORT_ID, DEST_CITY_NAME

ORDER BY inbound_passengers DESC

LIMIT 10; -- For TOP 10 */

-- Examine the relationship between city population and airport passenger traffic.

/*

with outgoing_passengers as

(select ORIGIN_AIRPORT_ID, SUM(PASSENGERS) as TOT_passengers

from airport_data

group by 1),

incoming_passengers as

(select DEST_AIRPORT_ID, SUM(PASSENGERS) as TOT_passengers

from airport_data

group by 1),

all_airports as

(select distinct ORIGIN_AIRPORT_ID as airport_id, ORIGIN_CITY_NAME as city_name from
airport_data

union

select distinct DEST_AIRPORT_ID as airport_id, DEST_CITY_NAME as city_name from
airport_data

),

passenger_traffic as

(select substring(aa.city_name,1,instr(aa.city_name,',') - 1) as city_name

, op.TOT_passengers + ip.TOT_passengers as total_passenger_travelling

from all_airports as aa

left join outgoing_passengers as op on aa.airport_id = op.ORIGIN_AIRPORT_ID

left join incoming_passengers as ip on aa.airport_id = ip.DEST_AIRPORT_ID

where op.TOT_passengers + ip.TOT_passengers is not null

order by total_passenger_travelling desc)

```
select cp.city_name, cp.Population, pt.total_passenger_travelling
from passenger_traffic as pt join city_population as cp on pt.city_name = cp.city_name
where Population is not null and total_passenger_travelling is not null
and Population <> 0
order by population asc */
```